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Associations of per- and polyfluoroalkyl substances with human milk metabolomic profiles in a rural North American cohort

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Background: Per- and polyfluoroalkyl substances (PFAS) are a class of persistent synthetic chemicals that are found in human milk and are associated with negative health effects. Research suggests that PFAS affect both lactation and the human metabolome.

Methods: We measured perfluorooctanoate (PFOA) and perfluorooctane sulfonate (PFOS) in the milk of 425 participants from the New Hampshire Birth Cohort Study using liquid chromatography-tandem mass spectrometry (LC-MS/MS). A nontargeted metabolomics assay was performed using LC with high-resolution MS, and metabolites were identified based on in-house database matching. We observed six metabolic profiles among our milk samples using self-organizing maps, and multinomial logistic regression was used to identify sociodemographic and perinatal predictors of these profiles, including infant sex, parity, participant body mass index, participant age, education, race, smoking status, gestational weight gain, and infant age at time of milk collection.

Results: Elevated PFOA was associated with profiles containing higher amounts of triglyceride fatty acids, glycerophospholipids and sphingolipids, and carnitine metabolites, as well as lower amounts of lactose and creatine phosphate. Lower concentrations of milk PFOS were associated with lower levels of fatty acids.

Conclusion: Our findings suggest that elevated PFOA in human milk is related to metabolomic profiles consistent with enlarged milk fat globule membranes and altered fatty acid metabolism. Further, our study supports the theory that PFAS share mammary epithelial membrane transport mechanisms with fatty acids and associate with metabolic markers of reduced milk production.

Key words: Per- and polyfluoroalkyl substances; Human milk; Metabolomics; Triglyceride fatty acids; Milk fat globule membrane; Lactation; Self-organizing maps

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Data are available through the Human Health Exposure Analysis Research Harmonized Data Portal https://hadatac-web.mssm.edu/hadatac. Human milk metabolomics data are available at the NIH Common Fund's National Metabolomics Data Repository (NMDR) website, the Metabolomics Workbench, https://www.metabolomicsworkbench.org, where it has been assigned Project ID ST001692.

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Introduction

Per- and polyfluoroalkyl substances (PFAS) are a class of over 12,000 synthetic chemicals that make consumer and industrial products water and oil resistant.[1] Due to their chemical composition, many resist degradation and can persist in the environment and in our bodies for decades.[2] PFAS are present in human milk in proportion to maternal serum PFAS concentrations, suggesting that breastfeeding is an important route of exposure for infants.[3–8] Neonatal PFAS exposure is associated with numerous negative health effects including reduced weight gain and increased liver weight.[9] Risk-benefit analyses of other chemicals present in human milk suggest that the benefits of breastfeeding outweigh the potential risk of chemical exposure, but such analyses are limited for PFAS.[10–13]

PFAS have been associated with shorter breastfeeding duration across geographically distinct and socioeconomically diverse populations and even among individuals with only background levels of exposure.[14–18] The mechanism for this association is unclear, although animal studies have suggested numerous mechanisms by which PFAS affect mammary gland development, maturation, and the process of lactation.[19–27] One underexplored mechanism for the association between PFAS exposure and shorter breastfeeding duration is the potential associations between individuals' PFAS exposure and the content or quality of the milk they produce.

What this study adds:
Our study is among the first to describe both per- and polyfluoroalkyl substances and metabolomics in human milk. We found associations between elevated perfluorooctanoate in milk and metabolites consistent with less nutritious milk and lower milk production.

Human milk is made of a wide range of bioactive components. Nearly 50% of the total energy content of human milk is fat, with 95%–99% of the fatty acids within triglyceride structures.[28] Fatty acids are passed into human milk from the lactating person's adipose stores and diet or synthesized by the mammary gland.[29,30] Epidemiologic studies have found that PFAS are associated with changes in lipid composition among adults, with alterations observed in cholesterol, apolipoproteins, fatty acids, and phospholipids concentrations, suggesting that PFAS play a role in lipid metabolism in the human body.[31–34] Further, research has shown associations between levels of PFAS in cord blood and changes in the human milk lipidome.[35,36] However, no studies have evaluated the relationship between PFAS concentrations in human milk and the metabolic composition of milk including both lipid and nonlipid components that provide vital contributions to growth, development, and energy for infants.

Our study sought to expand on previous lipidomics research to explore whether and how exposure to PFAS, as measured in human milk, affect the human milk metabolome and specifically the nutritive quality of milk. To do this, we studied associations between perfluorooctanoate (PFOA) and perfluorooctane sulfonate (PFOS) concentrations and metabolomic profiles in human milk among participants in the New Hampshire Birth Cohort Study (NHBCS). A better understanding of how environmental exposures such as PFAS affect infant nutrition can help to inform future research and public health interventions to optimize the nutritive properties of human milk and support lactating people to successfully feed their infants.

Methods

Study population

The NHBCS is a rural, ongoing pregnancy cohort beginning in 2009 that has accrued over 3000 maternal–infant dyads. The NHBCS has been described elsewhere.[37] Briefly, a series of questionnaires and medical record reviews collected information on lifestyle and demographic factors, pregnancy and birth outcomes, and reproductive and lactation history. Biological samples, including human milk, were collected for analysis. The current study used data from a subset of 425 participant–infant pairs who provided milk samples between 2014 and 2019. The study was approved by the Committee for the Protection of Human Subjects at Dartmouth College, and all participants provided informed consent.

Human milk collection

We collected pumped samples of milk from unsterilized bilateral breasts using a standardized protocol at the 6-week postpartum visit (mean 6.7 weeks [SD, 2.1]). Subjects provided a minimum of 18 ml and up to 80 ml of milk from each breast, with a median of 35 ml per breast. Samples were stored in the refrigerator at participants' homes for up to approximately 1 day and were brought in cold packs to the postpartum follow-up visit. Samples were processed within 24 hours of receipt and were stored at −80 °C until analysis.

Per- and polyfluoroalkyl substances in human milk

The analysis of milk was performed by the Minnesota Children's Health Exposure Analysis Research (CHEAR) Hub using protein precipitation followed by centrifugation, concentration, and analysis using liquid chromatography-tandem mass spectrometry (LC-MS/MS) to quantify concentrations of PFOA, PFOS, perfluorohexane sulfonate (PFHxS), perfluorononanoate, perfluorodecanoate, perfluoroundecanoate, perfluorobutane sulfonate, perfluoroheptanoate, perfluoroheptane sulfonate, and perfluorohexanoate. This method has been described elsewhere.[38] Accuracy for the method, indicated by % recovery, was determined through matrix spikes (n = 8) and ranged from 72% to 103%. Precision for the method, indicated by % relative standard deviation (RSD), was determined by repeated measurements of in-house prepared quality control (QC) materials (n = 20) 3%–18%. The limit of quantification (LOQ) ranged from 0.01 to 0.025 ng/ml for the analytes and values <LOQ were imputed to LOQ/√2.[39]

Human milk metabolomics measurement

Measurement of human milk metabolome was performed by the CHEAR/Human Health Exposure Analysis Research Laboratory Hub at the Department of Environmental Medicine and Public Health at the Icahn School of Medicine at Mount Sinai using validated methods.[40] Prior to sample preparation, milk samples were thawed on ice and vortexed to prevent separation of lipid-rich and aqueous phases. Extracts were prepared by treating 75 µl of milk with 150 µl of LC-MS-grade acetonitrile containing a series of 16 ^{13}C-labeled internal standards, vortexed, and allowed to equilibrate at 4 °C for 30 minutes. The extract was then centrifuged, and 50 µl of extract was transferred to an LC autosampler vial containing 50 µl of water and placed in a refrigerated autosampler until analysis.

Sample extracts were analyzed twice using an ultra-high performance LC equipped with dual pumps for both C18-positive and C18-negative analysis interfaced to a Thermo Scientific Vanquish Duo LC interfaced to a Thermo Scientific Q-Exactive HFX with electrospray ionization source operated in positive or negative mode. For C18-positive separation, 5 µl of the sample was injected onto C18 columns (50 mm Å~ 2.1 mm, 5µm particle size, Higgins Analytical Inc, Mountain View, CA) maintained at 30 °C. Separation occurred using mobile phase A consisting of water with 0.1% formic acid and mobile phase B consisting of acetonitrile with 0.1% formic acid. The flow rate was held at 0.4 ml/minute for 1.5 minutes and increased to 0.5 ml/minute. Solvent gradients were as follows: 85% solvent A, hold for 1.5 minutes; linear decrease to 5% solvent A at 5 minutes; hold for 2.5 minutes, for a total run time of 7.5 minutes. Data were acquired with a mass range of 85–1275 mass-to-charge ratio (m/z). The same process was used for C18-negative separation, with the substitution of water with 10 mM ammonium acetate for mobile phase A and acetonitrile alone for mobile phase B.

The high-resolution mass spectrometer was operated in full scan mode at 120,000 resolution and m/z range of 85–1275. Quality assurance and quality control (QAQC) were completed at the end of each batch (every 24 hours). Data were qualified using a five-step procedure that includes record keeping for study information (name, principal investigator information, date samples received, number of samples, number of batches, length, internal study ID, species, matrix, preservative, freeze/thaws prior to arrival), sample preparation record keeping (technician, sample preparation workflow, pooled sample batch, internal standards, number of QC samples), chromatographic QAQC (method information, solvents, run length, column and identifier information, injection volume, extracted ion chromatogram of representative metabolites for peak shape evaluation, retention time, and deviation), mass spectrometry QAQC (instrument name, tune file, polarity and resolution, median RSD for ^{13}C-labeled internal standards [same used for chromatography QAQC], RSD for internal standards for all samples in batch, RSD for analytes in pooled reference samples), and overall study QAQC (RSD and intensity distribution for common metabolites and internal standards across all batches, distribution of the

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sum of feature intensity across all samples and batch, intensity of analytes in method blanks). QAQC analysis was completed daily using an automated workflow combining TraceFinder with report generation by HTML and was verified by at least two different individuals. Untargeted feature tables were generated by converting .RAW files to mzXML and extracted using apLCMS at five different peak detection parameters. The resulting tables were merged using xMSanalyzer to remove redundant peaks, filtered to remove signals detected in <10% of samples, and batch-corrected using ComBAT.[41] Uniquely detected ions consisted of m/z, retention time, and ion abundance, referred to as m/z features. Untargeted feature tables were evaluated for batch effects before and after correction by comparing principal component (PC) scores for PC1, PC2, and PC3.

Metabolites were identified based on in-house database matching considering retention time, accurate mass, and MSMS matching (when available) with pure standards analyzed under the same conditions.[40]

Statistical analysis

We used the following to describe our study subsample: infant sex, parity (primiparous or multiparous), prepregnancy body mass index (BMI), participant age at time of enrollment, participant education, race/ethnicity as a proxy for the experience of structural racism, smoking status, gestational weight gain, and infant age at time of milk collection. We used only the results for milk concentrations of PFOS and PFOA because these two chemicals had the fewest number of samples <LOQ and represented the highest concentrations in our milk samples (Table S1; http://links.lww.com/EE/A310). Details of the observed PFAS milk concentrations have been previously described.[38]

Metabolomics data were paired to superclass information for metabolites from the Polyomics integrated Metabolomics Pipeline.[42] Given the study's hypothesis that PFAS influences the nutritional quality of human milk, we focused our analysis on common and well-known metabolites (160 of the 201 measured metabolites, see Table S2; http://links.lww.com/EE/A310 for metabolite names) and on metabolite ratios and sums that reflect the nutritional value of milk (Table S2; http://links.lww.com/EE/A310). These included the ratio of total saturated fatty acids (SFAs) to polyunsaturated fatty acids (PUFAs); omega-3 (n-3) fatty acids to omega-6 (n-6) fatty acids; palmitoleic acids (C16:1) to palmitic acid (C16:0); docosahexanoic acid (DHA, C22:6[n-3]) to arachidonic acid (ARA, C20:4[n-6]); and ARA to dihomo-gamma-linoleic acid (DGLA, C20:3[n-6]), all of which have been associated with alterations in metabolic health and inflammation.[43–48] We examined total polar lipids (sphingomyelins plus phospholipids) as a proxy for the presence and size of milk fat globule membranes in human milk. We included the sum of short- and medium-chain fatty acids (SCFA and MCFA), as these are produced by the mammary gland, and their production may be affected if mammary gland development or function has been affected by PFAS exposure.[14]

We defined profiles reflecting correlated metabolites and ratios measured in human milk using the self-organizing map (SOM) approach as previously described.[49–51] Briefly, this unsupervised clustering approach spatially plots correlated metabolites onto a one- or two-dimensional array (i.e., the "map") by combining nonlinear projection and cluster methods. The method involves an iterative learning process, where for our study, participants' multidimensional human milk metabolomic profiles were spatially compared to one another by differences in Euclidean distance (D):[50,51]

$$D\left(Z_i, Z_{i*}\right) = \left[\sum_{j=1}^{p}\left(Z_{ij} - Z_{i*j}\right)^2\right]^{1/2}$$

where (Z_i, Z_{i^*}) represents a pair of observations for the ith and i^*th participants, Z_{ij} and Z_{i^*j} are the ith and i^*th values of outcome j, and p is the total number of human milk metabolomic profiles. All outcomes used in analyses were continuous and standardized (mean = 0, SD = 1).

The SOM method involves multiple steps. First, multidimensional scaling was used to visually explore boundaries for how the data clustered. Next, we identified the number of clusters using several measures: the Calinski–Harabasz Index, which compares the ratio between the within-class variance and the between-class variance as a function of the number of classes, average silhouette width, R^2, and Akaike information criterion as described by Pearce et al.[50,51] (Figure S1; http://links.lww.com/EE/A310). Using these variance ratios we identified the number of SOM profiles by examining the agreement across statistics; if there are true k groupings in the data then the valid class solution will present itself with a shift (i.e., elbowing) in the evaluation statistic that can be visualized (Figure S1; http://links.lww.com/EE/A310).[50] SOM profiles were displayed graphically, and we visually assessed correlations to describe relevant patterns. Our intention of profile identification using unsupervised methods was exploratory and may inform hypothesis generation for further research.[52]

We used unadjusted multinomial logistic regression to assess different predictors of human milk metabolomic profiles compared to a natural referent category (profile 5). These independent predictors included: infant sex (male/female), parity (primiparous/multiparous), participant prepregnancy BMI (kg/m^2), participant age at enrollment (years), infant age at time of milk collection (days), and concentration of PFOA and PFOS in human milk (ng/ml). Additionally, associations between human milk metabolomic profile membership and PFAS human milk concentrations were adjusted for significant predictors found in our data (infant sex, age of child when milk sample was collected, gestational weight gain, and parity). Sensitivity analyses were conducted exploring these relationships using different reference groups and comparing with results from the main analysis (Tables S3 and S4; http://links.lww.com/EE/A310). Results are presented as medians with interquartile range (IQR) and odds ratios (OR) with 95% confidence intervals (CIs).

Results

Of the 425 participants with human milk samples, most tended to have some college education or more (78.7%), identified as non-Hispanic White (94%), and had never smoked (92%) (Table 1). On average, participants had a normal prepregnancy BMI. Mean age at time of enrollment was 32.2 years (SD, 4.5). The median level of PFOS in milk was 0.03 ng/ml (IQR: 0.02–0.04) and PFOA was 0.02 ng/ml (IQR: 0.01–0.03). Metabolite correlations are described in Figure S2; http://links.lww.com/EE/A310.

Metabolomic patterns were used to stratify participants into six groups identified by applying SOM, which included 3–189 participants in each group (Figure 1). Each node of the SOM represents a human milk metabolomic profile with no minimum threshold for node size, in which higher-than-average concentrations of each metabolite radiate from the center of each profile, while lower-than-average concentrations are represented in Figure 2. These profiles are arranged by hierarchical clustering, where adjacent profiles are more similar to one another than those farther away from one another. Bar plots for all data by profile are shown in Figure 2.

We observed several visual trends in the metabolic profiles of each group with a focus on common and well-known metabolites (Figure 1 and 2; Table S2; http://links.lww.com/EE/A310). Group 5 (n = 127) consisted of participants with average human milk metabolomic concentrations. Group 4 (n = 189) had individuals with just below-average concentrations for all

Table 1.

Characteristics of participants in the New Hampshire Birth Cohort Study by human milk metabolomic profile (N = 425)

Characteristic	All participants (N = 425) n (%) or mean (SD)	Profile 1 (n = 3) n (%) or mean (SD)	Profile 2 (n = 32) n (%) or mean (SD)	Profile 3 (n = 44) n (%) or mean (SD)	Profile 4 (n = 189) n (%) or mean (SD)	Profile 5 (n = 127) n (%) or mean (SD)	Profile 6 (n = 30) n (%) or mean (SD)
Infant sex							
Male	218 (51.3%)	1 (33.3%)	18 (56.2%)	19 (43.2%)	94 (49.7%)	74 (58.3%)	12 (40.0%)
Female	207 (48.7%)	2 (66.7%)	14 (43.8%)	25 (56.8%)	95 (50.3%)	53 (41.7%)	18 (60.0%)
Parity							
Primiparous	178 (43.7%)	2 (66.7%)	13 (40.6%)	27 (61.4%)	68 (38.0%)	53 (43.4%)	16 (59.3%)
Multiparous	229 (56.3%)	1 (33.3%)	19 (59.4%)	17 (38.6%)	111 (62.0%)	69 (56.6%)	11 (40.7%)
Highest attained education							
Less than high school	32 (8.3%)	0 (0%)	6 (22.2%)	3 (7.7%)	12 (7.0%)	6 (5.1%)	5 (17.2%)
Graduated high school	50 (13.0%)	1 (50.0%)	3 (11.1%)	11 (28.2%)	15 (8.8%)	19 (16.2%)	1 (3.4%)
Some college	155 (40.3%)	1 (50.0%)	9 (33.3%)	11 (28.2%)	80 (46.8%)	42 (35.9%)	12 (41.4%)
Completed higher education	148 (38.4%)	0 (0%)	9 (33.3%)	14 (35.9%)	64 (37.4%)	50 (42.7%)	11 (37.9%)
Race/ethnicity							
Non-Hispanic White	401 (94.4%)	3 (100%)	31 (96.9%)	43 (97.7%)	179 (94.7%)	117 (92.1%)	28 (93.3%)
Non-White	24 (5.6%)	0 (0%)	1 (3.1%)	1 (2.3%)	10 (5.3%)	10 (7.9%)	2 (6.7%)
Smoking status							
Never	354 (83.2%)	1 (33.3%)	25 (78.1%)	37 (84.1%)	159 (84.1%)	107 (84.2%)	25 (83.3%)
Former	17 (4.0%)	2 (66.7%)	3 (9.4%)	0 (0.0%)	8 (4.2%)	4 (3.2%)	0 (0.0%)
Current	17 (4.0%)	0 (0.0%)	1 (3.1%)	2 (4.5%)	5 (2.7%)	6 (4.7%)	3 (10.0%)
Missing[a]	35 (8.2%)	0 (0.0%)	3 (9.4%)	5 (11.4%)	17 (9.0%)	10 (7.9%)	2 (6.7%)
Prepregnancy BMI	25.7 (6.1)	22.3 (2.3)	26.4 (8.2)	26.1 (6.6)	25.7 (6.1)	25.4 (5.3)	26.2 (6.3)
Enrolment age (years)	32.2 (4.5)	27.1 (5.5)	31.8 (5.3)	31.2 (4.1)	32.4 (4.2)	32.4 (4.9)	32.2 (4.0)
Gestational weight gain (pounds)	31.4 (14.1)	34.0 (7.1)	27.2 (13.7)	27.2 (12.2)	32.1 (14.9)	33.3 (13.8)	30.2 (11.9)
Milk age (collection after birth weeks)	6.7 (2.1)	6.1 (0.7)	5.9 (1.3)	7.8 (3.0)	6.5 (1.3)	6.9 (2.6)	6.7 (2.2)
PFOS in human milk concentration (ng/ml)	0.03 (0.02)	0.01 (0.005)	0.04 (0.02)	0.03 (0.02)	0.03 (0.02)	0.03 (0.02)	0.03 (0.01)
PFOA in human milk concentration (ng/ml)	0.02 (0.01)	0.01 (0.006)	0.03 (0.02)	0.02 (0.01)	0.02 (0.01)	0.02 (0.01)	0.02 (0.01)

[a]Missing participants for variables: education (n = 38); gestational weight gain (n = 41).

metabolites. Group 3 (n = 44) had among the highest free fatty acid concentrations of the total group, including total SCFA and MCFA, as well as higher levels of retinoic acid and vitamin A, and lower levels of lactose. While slightly low in fatty acid concentration, Group 2 (n = 32) had elevated levels of carnitine metabolites and glycerophospho- and sphingolipids, including lysophosphatidylcholines and sphingomyelins, and total polar lipids, but lower levels of lactose. Group 6 (n = 30) was low in fatty acids and sphingolipids but had elevated levels of saccharides including lactose and amino acids including creatine phosphate. Group 1 had only three members and was therefore excluded from subsequent regression models.

In unadjusted models, participants in groups 3 and 6 were less likely to have male infants as compared to group 5 participants (OR [95% CI]: 0.54 [0.27, 1.08]; 0.48 [0.21, 1.07], respectively) (Table 2). Group 3 participants were twice as likely to be primiparous as compared to group 5 participants (OR: 2.07 [95% CI: 1.02, 4.18]). Participants in groups 2 and 3 had slightly lower odds of increased gestational weight gain than participants in group 5 (OR [95% CI]: 0.97 [0.94, 0.99]; 0.97 [0.94, 0.99], respectively).

After adjusting for additional covariates including infant sex, age of child when milk sample was collected, gestational weight gain, and parity, we modeled the association between PFOS or PFOA concentrations in human milk and SOM group membership using multinomial logistic regression (Figure 3; Table 2). Participants in group 4 were less likely to have elevated levels of PFOS in their milk (OR [95% CI]: 0.65 [0.44, 0.95]) as compared to participants in group 5, and participants in groups 2 and 3 were more likely to have elevated levels of PFOA in their milk (OR [95% CI]: 2.60 [1.15, 5.80] 1.70 [0.87, 3.44], respectively). In sensitivity analyses, we evaluated the robustness of our findings by changing the referent group, and we compared the results to the main findings. We used the following referent groups: the highest n-3/n-6 fatty acid ratio, the group with the largest sample size (i.e., statistical power), and the group with overall average metabolomic profile as compared to all data. Changes in the referent group showed similar results to the main analysis (Figures S3 and S4; http://links.lww.com/EE/A310).

Discussion

In the NHBCS, we observed five major nutritional profiles of 160 common and well-known metabolites in human milk and associations between milk concentrations of PFOS and PFOA and these profiles. While PFOS and PFOA concentrations in our milk samples were overall commensurate with other background exposure cohorts,[38] higher levels of PFOA were found in the milk of participants in group 2, characterized by elevated carnitine metabolites and glycerophospho- and sphingolipids and lower triglyceride fatty acids and lactose, and in group 3, characterized by elevated triglyceride fatty acids, retinoic acid, and vitamin A and lower lactose. Lower levels of PFOS were found in the milk of participants in group 4, characterized by low triglyceride fatty acids. We did not find any association between PFAS concentrations and the metabolomic profile of milk in group 6, characterized by lower glycerophospho- and sphingolipids and elevated amino acids and saccharides, or with ratios of SFA/PUFAs, n-3/n-6 fatty acids, C16:1/C16:0, DHA/ARA, or ARA/DGLA in any group.

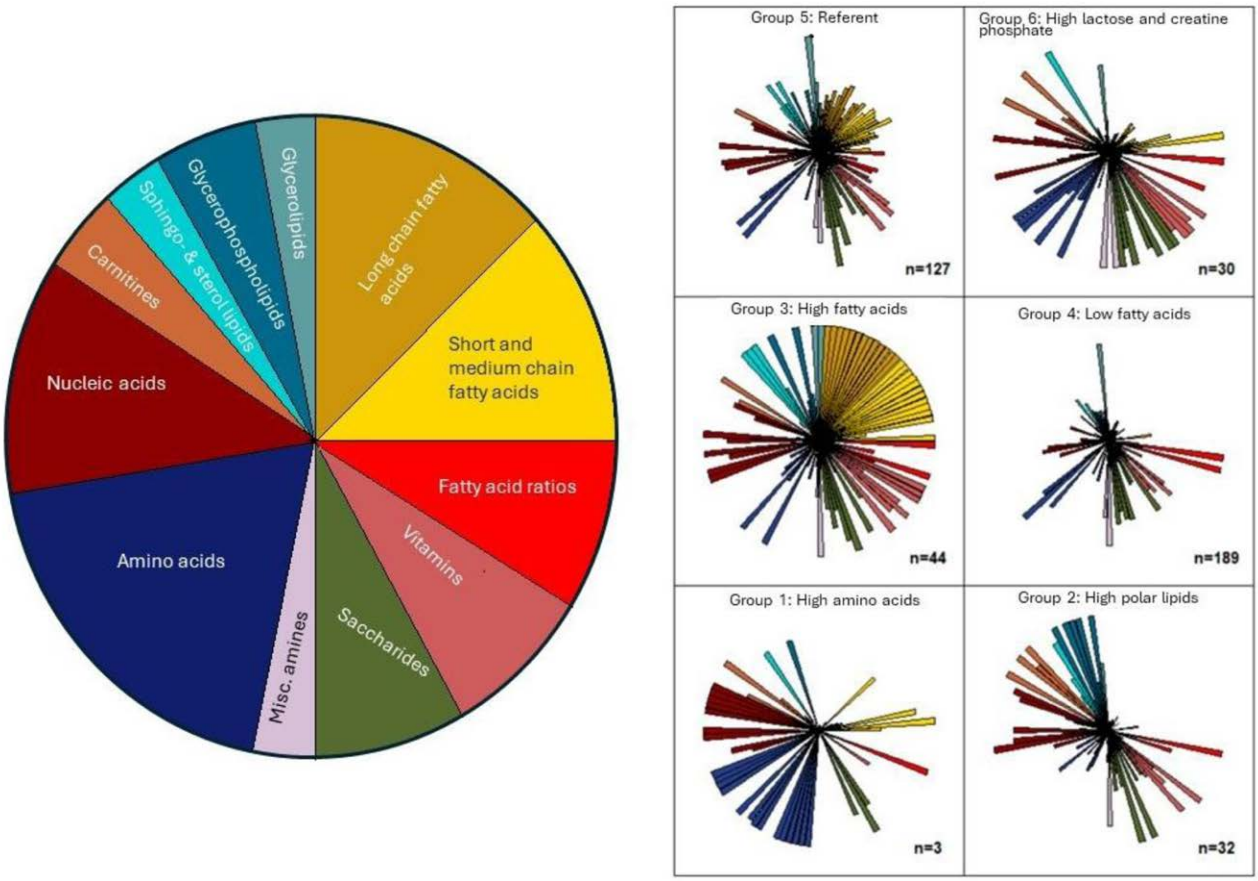


Figure 1. Self-organizing map (SOM) of human milk metabolomic profiles of the New Hampshire Birth Cohort Study (N = 425 participants). Metabolic concentrations mean centered (mean = 0) and scaled (SD = 1). Bar height presents the abundance of metabolite within the group of participants. Color indicates the class of metabolites to which each metabolite belongs.

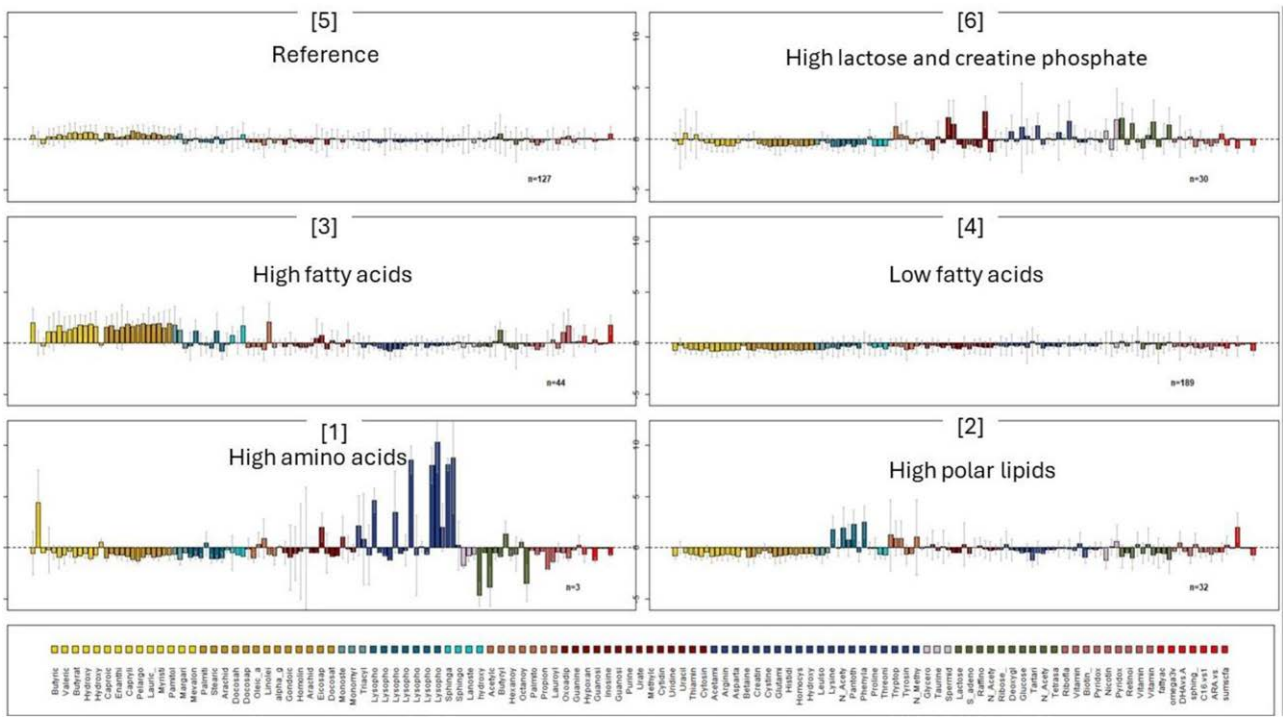


Figure 2. Mean-centered (SD) concentrations of metabolites in human milk metabolomic profiles of the New Hampshire Birth Cohort Study (N = 425), where *x*-axis represents the concentration of metabolites in milk relative to the mean concentration of the overall sample (N = 425) and the *y*-axis represents individual metabolites as depicted in the key below.

Table 2.

Predictors of human milk metabolomic profiles using unadjusted multinomial logistic regression from participants of the New Hampshire Birth Cohort Study (n = 422)

Predictor	N missing	Reference (group 5, n = 127) n or % Ref	High polar lipids (group 2, n = 32) % or mean (SD) OR (95% CI)	High fatty acids (group 3, n = 44) % or mean (SD) OR (95% CI)	Low fatty acids (group 4, n = 189) % or mean (SD) OR (95% CI)	High lactose and creatine phosphate (group 6, n = 30) % or mean (SD) OR (95% CI)
Male infant	0	58.5%	56.2%	40.5%	49.5%	41.9%
		Ref	0.92 (0.42, 2.01)	0.54 (0.27, 1.08)	0.71 (0.45, 1.12)	0.48 (0.21, 1.07)
Primiparity	17	43.4%	40.6%	61.4%	38.0%	59.3%
		Ref	0.89 (0.40, 1.96)	**2.07 (1.02, 4.18)**	0.80 (0.50, 1.27)	1.89 (0.81, 4.42)
BMI	0	25.5 (5.3)	26.4 (8.2)	26.1 (6.6)	25.7 (6.1)	26.2 (6.3)
		Ref	1.02 (0.96, 1.09)	1.02 (0.96, 1.08)	1.00 (0.97, 1.05)	1.02 (0.96, 1.09)
Enrolment age	0	32.4 (4.9)	31.8 (5.3)	31.1 (4.2)	32.3 (4.1)	32.5 (4.1)
		Ref	0.97 (0.89, 1.06)	0.94 (0.87, 1.01)	1.00 (0.95. 1.05)	0.99 (0.90, 1.08)
Gestational weight gain	41	33.1 (14.0)	26.7 (13.6)	27.8 (11.7)	32.1 (15.0)	30.0 (11.7)
		Ref	**0.97 (0.94, 0.99)**	**0.97 (0.94, 0.99)**	0.99 (0.98, 1.01)	0.98 (0.95, 1.01)
Milk age	0	6.9 (2.6)	5.9 (1.3)	7.8 (3.0)	6.5 (1.3)	6.7 (2.2)
		Ref	**0.50 (0.33, 0.78)**	1.11 (0.99, 1.25)	0.88 (0.77, 1.00)	0.97 (0.79, 1.19)
PFOS concentration[a]	0	0.03 (0.02)	0.04 (0.02)	0.03 (0.02)	0.3 (0.02)	0.03 (0.01)
		Ref	1.55 (0.80, 3.00)	1.49 (0.84, 2.67)	**0.64 (0.43, 0.94)**	0.88 (0.45, 1.72)
PFOA concentration[a]	0	0.02 (0.02)	0.03 (0.02)	0.02 (0.01)	0.02 (0.01)	0.02 (0.01)
		Ref	**2.15 (1.12, 4.10)**	**1.78 (1.00, 3.16)**	0.82 (0.55, 1.20)	1.39 (0.71, 2.70)

Bold values indicate a significant association.

[a]Results reflect natural log transformation of PFOS and PFOA.

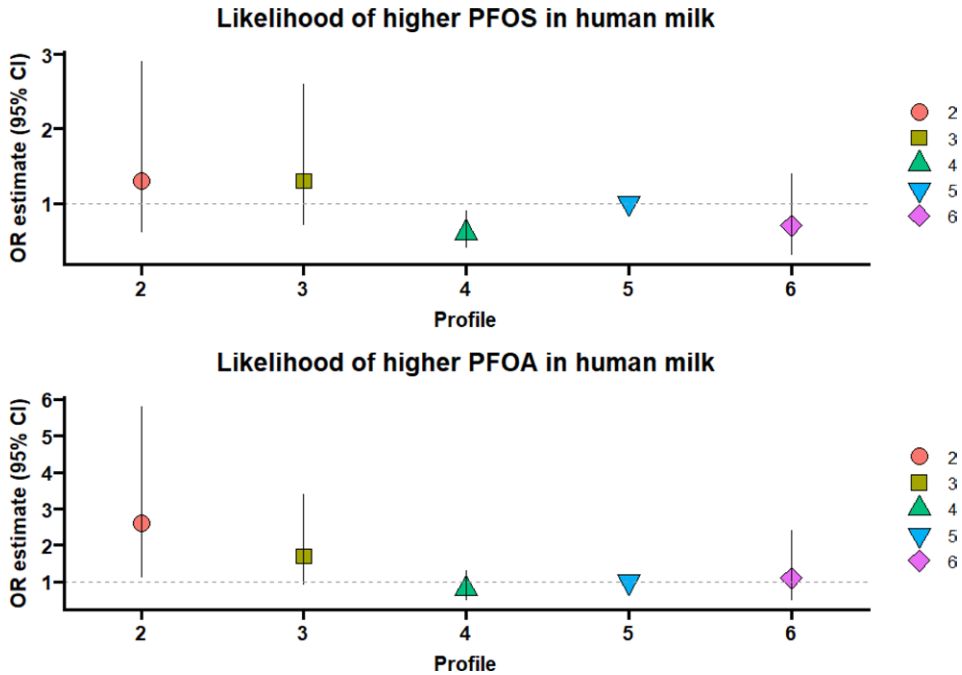


Figure 3. Odds ratios (95% CIs)[a] between human milk metabolomic profile membership[b] and natural log of PFAS concentrations in human milk, as compared to the reference group, the average human milk metabolomic profile (profile 5) in the New Hampshire Birth Cohort Study (n = 422). [a]Adjusted for infant sex, age of child when milk sample was collected, gestational weight gain, and parity. [b]Group numbers represent the following: group 2: high polar lipids; group 3: high fatty acids; group 4: low fatty acids; group 5: reference group; group 6: high lactose and creatine phosphate.

Parity, infant sex, and gestational weight gain were associated with fatty acids in patterns in milk consistent with existing research. Primiparity was associated with higher levels of all fatty acids (profile 3), which may be due to the relative depletion of the lactating person's fatty acid stores over the course of multiple births and lactation periods.[53–56] Increased levels of lactose and creatine phosphate (profile 6), considered markers for milk production,[57] were found among participants with female infants. While animal data suggest certain sex-specific differences in milk composition, human data on this topic are conflicting and sparse.[58] Decreased gestational weight gain was associated both with a lower fatty acid profile (group 2) and a higher fatty acid profile (group 3), although there were no specific patterns relating gestational weight gain to putative healthy or unhealthy fatty acids or ratios. Triglyceride fatty acids compose the majority of the fat content of human milk,[29,30] and the quantity and mixture of fatty acids to which an infant is exposed through human milk can have positive and negative health effects.[59]

Several studies have shown associations between obesity and gestational weight gain and the fatty acid composition of milk; however, findings tend to be nuanced, which may account for our

contradictory findings. For example, a Norwegian study found an association between pregnancy weight gain and concentration of monounsaturated fatty acids (MUFAs) in human milk,[53] while a Canadian study found that maternal obesity was associated with higher SFAs, higher MCFAs, and lower MUFAs.[60] Similarly, the Finnish STEPS (Steps to the Healthy Development and Well-being of Children) study showed that the milk of overweight mothers contained higher SFAs, lower n-3 fatty acids, a lower ratio of unsaturated to SFAs, and a higher n-6/n-3 fatty acid ratio, overall suggesting less nutritional milk.[61]

The nutritional content of human milk is affected by a myriad of factors, including the lactating person's diet, geographic location, lifestyle, time of feed or day, and infant age.[47,57,62–65] For these reasons, it is impossible to define an optimal human milk nutritional profile, although certain metabolites and ratios of metabolites are known to support infant growth and development. Our findings suggest that exposure to PFAS may be associated with some less nutritious milk metabolites (e.g., SFAs, MUFAs, n-6 fatty acids, and excesses of glycerophospholipids, sphingolipids, and carnitine) and suboptimal milk production (e.g., lower levels of lactose and creatine phosphate), although there were no associations between PFOA or PFOS and any of the fatty acid ratios considered nutritional markers (e.g., SFA/PUFA, n-3/n-6 fatty acids, C16:1/C16:0, DHA/ARA, and ARA/DGLA).

Interestingly, we found an almost doubled odds of increased levels of PFOA in the milk of group 3 participants (high fatty acids), although the 95% CI for this association contained the null. We found that participants in group 4 (lower fatty acids) had nearly half the odds of having elevated PFOS in their milk. Similarly, a 2021 study of 44 Finnish mothers found that early milk samples (2–4 days postnatal) among mothers with the highest PFAS levels had elevated levels of SFAs and MUFAs and decreased levels of PUFAs, although there was no longer a difference in milk samples collected at 3 months postpartum.[36] Among our study participants, group 3, with elevated levels of PFOA, had almost twice the concentration of SCFA and MCFA acids in their milk as reference group 5 participants. Most SCFAs and MCFAs are produced by the mammary gland, and these findings together may reinforce mechanistic studies suggesting that exposure to PFAS alters mammary gland function.[14] However, group 3 participants had elevated levels of all fatty acids in their milk, including PUFAs, as compared to group 5, and so mammary gland dysfunction alone may not account for our findings.

The Finnish study found an inverse relationship between PFAS exposure and DHA in early milk samples, which has numerous benefits for infant growth and development.[36] We found that participants in groups 2 and 3 had slightly elevated n-3 to n-6 PUFA ratios, suggesting an opposite association. Differences in findings may be accounted for by differences in the timing of milk collection (2–4 days in the Finnish Study versus ~6 weeks in our study), PFAS measurement (serum vs. milk), and dietary exposure to DHA, which may be higher in a Finnish population with a diet high in fish.[36]

Our finding of elevated PFOA in group 3 (high fatty acids) and decreased PFOS in group 4 (lower fatty acids) may further offer insight into how PFAS is transferred into human milk. Unlike other environmental toxicants that are lipophilic, PFAS largely exist in humans bound tightly to the protein albumin.[4] At present little is known about how PFAS are transferred from the lactating person's serum into human milk, although research notes their structural similarity to triglyceride fatty acids.[66] For this reason, some propose that PFAS are transported into milk in the same way as fatty acids: either transferred from albumin by active transport across the mammary epithelium or unbound via passive diffusion from serum.[67] However, the tight bond between albumin and PFAS raises a question as to whether PFAS is able to interact with fatty acid transporters to pass into human milk.[4] The co-existence of elevated PFOA and elevated levels of fatty acids in the milk of group 3 participants and the decreased PFOS and fatty acids in group 4 participants among our study cohort may lend support to the hypothesis that PFAS are actively transported in the same fashion as fatty acids. Future experimental research is needed to evaluate whether PFAS enter human milk via active transport mechanisms.

We found elevated levels of carnitine metabolites in group 2 participants, with elevated levels of PFOA. Carnitine metabolites are the byproducts of fatty acid beta-oxidation in humans, which is essential for the utilization of fatty acids for energy.[68] Their presence in human milk is less well characterized, although one study found maternal BMI to be associated with increased acylcarnitines in human milk at 6 months postpartum along with a milk metabolome pattern consistent with plasma metabolomic signatures in obesity and type 2 diabetes.[69] Increased levels of carnitine metabolites in milk may represent less nutritious milk, or they may represent shifts in the metabolic activity of the mammary gland.[70] Animal models have found that mice treated with PFOA had alterations in the genes related to fatty acid catabolism, which may explain the co-existence of higher levels of PFOA and carnitine metabolites, the byproducts of fatty acid catabolism in milk.[71]

Glycerophospho- and sphingolipids including lysophosphatidylcholines and sphingomyelins are often found in the milk fat globule membrane and are related to infant gut microbiome composition and cognitive development.[62] Vitamin A and retinol are also found in the milk fat globule membrane.[72] We found that groups 2 and 3 participants, with elevated PFOA exposure, had elevated levels of glycerophospho- and sphingolipids and vitamin A and retinol, respectively. These findings are similar to those found among Finnish mothers, where PFOS and PFHxS were associated with elevated levels of phosphatidylethanolamines (PE), and PFOA and PFOS were associated with a decreased phosphatidylcholines (PC)/PE ratio, suggesting larger milk fat globule membranes.[36] Our SOMs did not include PC or PE, but group 2 participants also had increased total sphingomyelins and phospholipids, which suggest similar findings regarding milk fat globule membrane size. Some mechanistic studies indicate that larger milk fat globule membranes are more difficult for infants to digest and lead to impaired growth and increased gut inflammation.[36,73] Epidemiologic studies have suggested that infant exposure to PFAS via human milk may increase infant gut biodiversity but alter the relative abundance of biologically active bacteria, possibly predisposing to infection.[74]

Lactose and creatine phosphate, both present in lower amounts in groups 2 and 3, are considered proxies for milk production.[57] Numerous studies have suggested that elevated levels of PFAS in a lactating person's serum or milk are associated with decreased breastfeeding duration.[14–18] Our findings suggest that a mechanism behind this association may be reduced milk volume in addition to altered milk nutrition, which may result in inadequate calories for infant growth and development. Further research into actual milk volumes produced is necessary to elucidate this hypothesis.

To our knowledge, our study is among the first to characterize the human milk metabolome in relation to PFAS exposure and is the first to do so in such a large, contemporary, well-characterized cohort. Further, all milk samples were collected during a short timeframe postpartum, meaning that changes in milk PFAS and metabolomic profiles due to time from birth are minimized. While the relative homogeneity of our cohort may limit external generalizability, it also means that variability in results due to other covariates is minimal, and as such, the NHBCS can form the basis for future hypotheses in this novel arena. While the NHBCS captures the exposure experience of a vulnerable population of rural pregnant people who use private water systems that may contain environmental contaminants above public drinking water regulatory levels, our sample's PFAS concentrations are consistent with levels of exposure

found in the general population. As such, these results may not be reflective of the experience of highly exposed rural populations, such as those affected by PFAS-contaminated biosolids used in agricultural settings in rural Maine.[75] Finally, our study explores only two of a class of thousands of PFAS chemicals, and future research may expand the number of PFAS studied.

In summary, in a subsample of participants from a rural birth cohort, we observed five nutritional profiles in human milk, including neutral, high fatty acid, high glycerophospho- and sphingolipids, low fatty acid, and high lactose and creatine phosphate. Elevated levels of PFOA were associated with high fatty acid levels, providing additional mechanistic support that PFAS are transferred into milk in the same manner as triglyceride fatty acids. Elevated PFOA levels were also associated with profiles suggesting less healthy milk fat globule membrane formation and decreased milk production. Future studies should explore PFAS associations with milk volume and further interrogate the mechanisms by which PFAS are transferred from the plasma of lactating individuals into human milk.

Conflicts of interest statement

The authors declare that they have no conflicts of interest with regard to the content of this report.

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